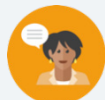


3.8 Analyzing Residuals

Lesson Introduction

 Benchmark	MA.912.DP.2.6 Given a scatter plot with a line of fit and residuals, determine the strength and direction of the correlation. Interpret strength and direction within a real- world context.		 Learning Targets	- I can determine whether a function is an appropriate fit for the data by analyzing residuals. - I can interpret the strength and direction of the correlation within a real-world context.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.912.DP.2.7 Compute the correlation coefficient of a linear model using technology. Interpret the strength and direction of the correlation coefficient.	
 Essential Questions	- How can you determine whether a function is an appropriate fit for the data by analyzing residuals? - How can you interpret the strength and direction of the correlation within a real-world context?			
 Lesson Narrative	In this lesson, students are introduced to residual values. Students will learn how to determine whether a function is an appropriate fit for the data by analyzing residuals. As part of the analysis, students will also interpret the strength and direction of the correlation within a real-world context. The lesson begins with exploration, before bringing the ideas together and connecting to previously learned content from the unit.			
 Component Summary	3.8.1 Warm-Up (~5 min.)	Consider strengths of linear associations in bivariate data <i>(SE p. 150)</i>	3.8.3 Guided Instruction (10-15 min.)	Introduction to analyzing residuals <i>(SE p. 152)</i>
	3.8.2 Exploration (10-15 min.)	Explore predicted vs actual values in bivariate data <i>(SE p. 151)</i>	3.8.4 Bring It Together (10-15 min.)	Summarize how to interpret the strength and direction of a line of fit <i>(SE p. 154)</i>
	3.8.5 Wrap-Up (~5 min.)	Coordinate plane reflection activity <i>(SE p. 155)</i>		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms</u> : N/A <u>Additional Glossary</u> : N/A		(Optional) Graphing utility	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse residuals with the actual data values. - Students may use an inappropriate line of best fit for the given data. - Students may misinterpret the strength and direction of an accurate line of fit.	- The learning activity series in this lesson is designed to bring students through progression of learning about residuals from developing a conceptual component to an analysis within context component. Consider going through these activities in order, prior to instruction, to better understand the lesson design and the student experience. - The 3.8.2 Exploration is a crucial point of this lesson. The conceptual foundation that this component lays should not be rushed. Allow students to explore and ask questions that can then be fully addressed through the 3.8.3 Guided Instruction and 3.8.4 Bring It Together.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display 3.8.2 Exploration provides students with an opportunity to explore predicted vs actual values as they conceptualize what residuals are. During this activity, student language is key. The Collect and Display routine is built into the activity by providing students the chance to first think about a response, and then to refine that response through additional activities.	MA.K12.MTR.1.1: Participate in Effortful Learning In 3.8.2 Exploration, students will collaborate to explore predicted vs actual data values from a scatter plot with a line fit to the data. Students will be able to ask questions that will help with solving the task. By working together, they can help and support each other when attempting to practice this new concept.
 Differentiate Instruction	Consider going digital with this lesson. There are several visuals that are used throughout the lesson. If students may need larger graphs, consider taking the images and putting them in a graphing utility that can display the image to students on their device (if available) to be able to explore the graph up close and zoom in where needed.	
Lesson Notes:		

3.8.1 Warm-Up: Bell Work

Component Overview: Students will consider four scatter plots to arrange them in an order from least closely fit to a line to most closely fit.



3.8.1 Warm-Up: Bell Work

Four scatter plots are shown.



At the conclusion of this activity, if time allows, have students share how they made their decisions. They could also talk about the direction of the relationship between those that are on the right side of the scale.

- Order the scatter plots from bivariate data that is "least closely fit to a line" to "most closely fit to a line" by writing the letter corresponding to each graph.

Least Closely Fit to a Line		Most Closely Fit to a Line	
D	B	A	C

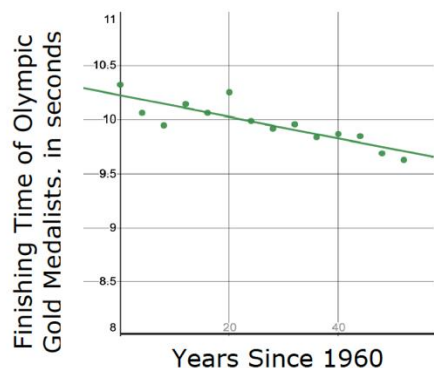
3.8.2 Exploration: Predicted vs. Actual Values

Component Overview: Students will explore the concept of predicted vs actual values using a real-world scenario and three, scaffolded questions. For this activity, students will need a partner.



3.8.2 Exploration: Predicted vs. Actual Values

The scatter plot below shows the finishing times for the Olympic gold medalist in the men's 100-meter (m) dash for the Olympic games since 1960.



The raw data used to construct the scatter plot is shown.

Year of Olympics	1960	1964	1968	1972	1976	1980	1984
Gold Medal Time (in seconds)	10.32	10.06	9.95	10.14	10.06	10.25	9.99
Year of Olympics	1988	1992	1996	2000	2004	2008	2012
Gold Medal Time (in seconds)	9.92	9.96	9.84	9.87	9.85	9.69	9.63

The equation of the line of best fit that models the relationship is $f(x) = 10.2217 - 0.0098x$, where f is the finishing time of the Olympic gold medalist in the men's 100-meter dash and x is the number of years since 1960. Work with your partner to complete the following using the information on the previous page.



During this activity, students are working towards conceptually understanding residual values. Be careful not to give away too much vocabulary and process steps in this exploration. This activity is designed to lead students through a scaffolded discovery of predicted values and actual values and what those mean.



This exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next two components.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group Bring It Together.

3.8.2 Exploration: Predicted vs. Actual Values (continued)

1. Consider the 2012 Summer Olympic games.
 - a. How many years after 1960 was the 2012 Summer Olympics?
52 years
 - b. Using the equation that models the relationship between years since 1960 and the Olympic gold medal finishing times, what is the predicted finishing time for the men's gold medalist in 2012, rounded to the nearest hundredth?
9.71 seconds
 - c. What was the actual finishing time for the men's gold medalist in 2012?
9.63 seconds
 - d. Complete the statement using your answers from Parts B and C.
The actual finishing time for the men's gold medalist in the 2012 Summer Olympics was

☒ faster
☐ slower

 than predicted using the line of best fit
by 0.08 seconds.
2. Identify one point on the scatter plot where the actual finishing time for the gold medalist was more than the model predicted.
 - a. Find the coordinates of the point.
Answers will vary. (20,10.25); (40,9.87)
 - b. Explain how you know the actual finishing time was more than the model predicted.
Answers will vary. I knew that the actual finishing time was more than the model predicted because the point was above the line of best fit.



For question 1 part d, students may struggle with selecting the correct answer with faster/slower in this context. Assist students with understanding that a lesser time is a faster finish.



For question 2, students may not understand the meaning of "model" referred to in the question stem. Ensure students understand the model is the equation or the line of best fit. This line "models" the relationship between the bivariate data.

3.8.2 Exploration: Predicted vs. Actual Values (continued)

3. Discuss with your partner whether the linear function appears to be a good fit for the data and how you know. Summarize your discussion.

Answers will vary. The linear function appears to be a good fit for the data because there is an approximately equal distribution of data points above and below the line, with some appearing to lie on the line. Also, the actual data values are not very far from the line being used to model the trend.

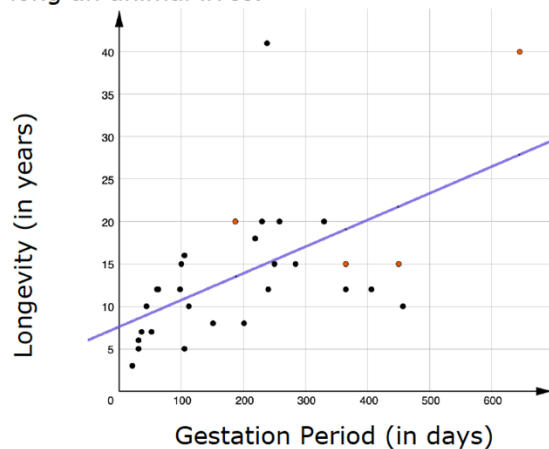
3.8.3 Guided Instruction: Analyzing Residuals

Component Overview: Students will be introduced to analyzing residuals of a real-world problem using a scaffolded, multipart, two-question series. For this activity, students will need a partner for discussion.



3.8.3 Guided Instruction: Analyzing Residuals

The scatter plot summarizes the average gestation period, in days, and average longevity, in years, for a sample of 30 animals as reported in *The World Almanac and Book of Facts* (McGeveran, 2006). A gestation period is the amount of time for which an animal is pregnant. Longevity refers to how long an animal lives.



The function $g(x) = 0.032x + 7.6$ is fit to the data.

- Consider the table shown with the x -values, actual y -values, predicted y -values, and residual values of the orange points on the scatter plot.

x -value	Actual y -value	Predicted y -value	Residual Value
187	20	13.584	6.416
365	15	19.28	-4.28
450	15	22	-7
645	40	28.24	11.76

- Discuss with your partner what you think a residual value is.



Add a layer of engagement by pulling “*The World Almanac and Book of Facts*” up online so students have a better contextual understanding of what this is and how the information was gathered from it.

Highlight the information for the terms “gestation period” and “longevity” so that students have a general understanding, so that it won’t be a barrier in their analysis of the data.



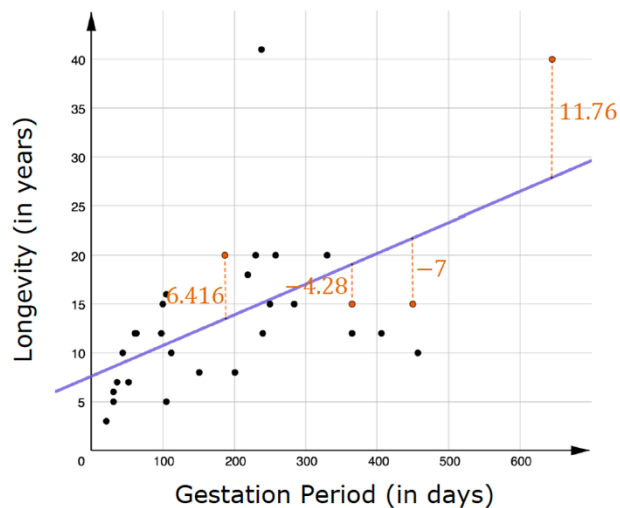
Consider providing students the opportunity to first discuss the context before engaging with question 1. Model how to mark the text to pull out critical information from the context and make connections across the different representations of the situations (i.e., the text, graph, and equation of the line of fit).



Consider allowing students to share out their responses to question 1 part a. These responses can be charted on paper or a whiteboard as a reference to initial thoughts on the term itself, and then refer to once they know the formal definition.

3.8.3 Guided Instruction: Analyzing Residuals (continued)

The scatter plot is shown, with the residual values of the four points from the table labeled.



b. Complete the statements.

A residual value is the difference,

- ☐ (actual y -value) - (predicted y -value) for a
☐ (actual y -value) + (predicted y -value)

given x -value in the data set. A residual value

- ☐ close to
☐ far from
- 0 indicates the actual data value is

close to the line of best fit.



To help students understand why the values are not adding consider asking the following questions:

- Why do you think that the predicted y -value would be subtracted from the actual y -value, and not added?
- What are you finding on the graph when you subtract the two values?

3.8.3 Guided Instruction: Analyzing Residuals (continued)

A positive residual value means the actual point lies

- ☐ on
- ☒ above
- ☐ below

the line of best fit. A negative residual

value means the actual point lies

- ☐ on
- ☐ above
- ☒ below

of best fit. A residual value of 0 means the actual

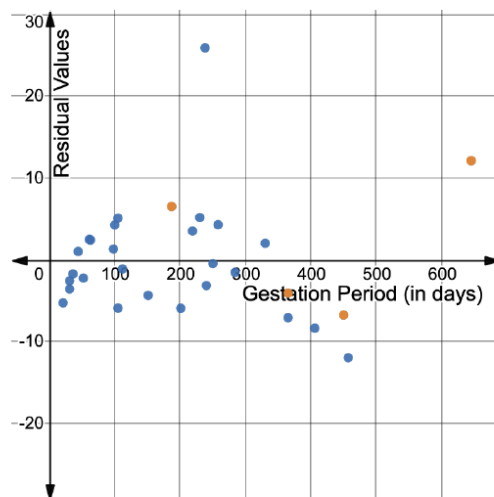
point lies

- ☒ on
- ☐ above
- ☐ below

the line of best fit.

Residuals are used to analyze the strength of a line's fit to the data. They can be represented on a residual plot like the one shown below.

Each point on the residual plot shown shows the residual value for a corresponding point on the scatter plot. The residual values explored in Question 1 are shown in orange.



For a deeper understanding beyond Quadrant 1, consider the following question:

- Would this statement still be true if the line of best fit and data were all found in Quadrant 4?



What do you notice about the data displayed in the graph?
What do you wonder?

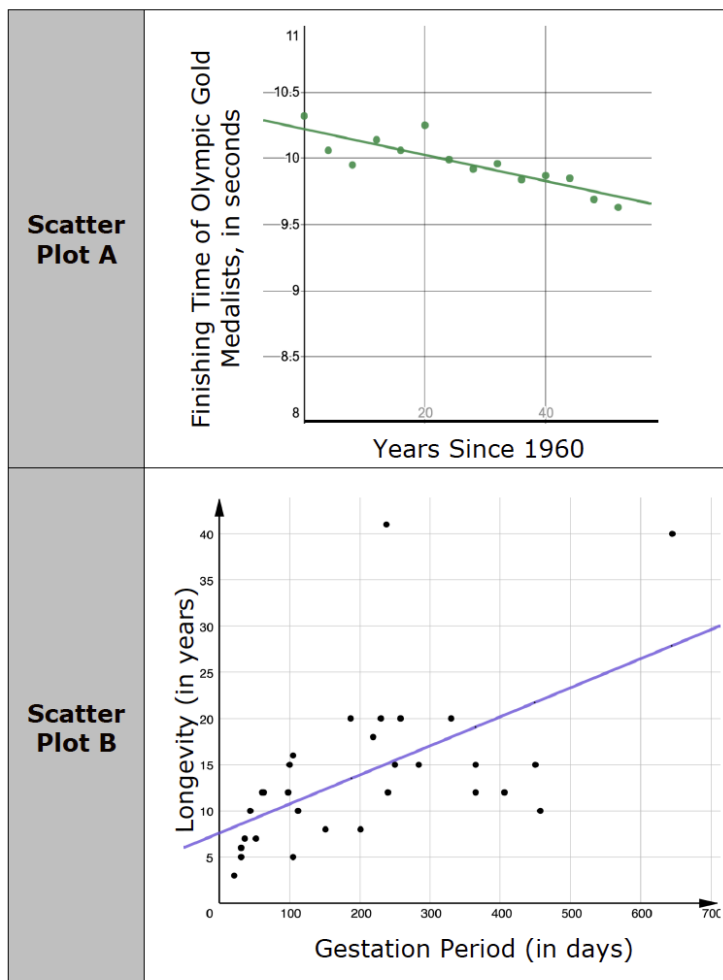
3.8.4 Bring It Together: Interpreting the Strength and Direction of a Line of Fit

Component Overview: Students will formalize what they've learned about residuals to interpret the strength and direction of a line of fit.



3.8.4 Bring It Together: Interpreting the Strength and Direction of a Line of Fit

The two scatter plots explored in this lesson are shown.



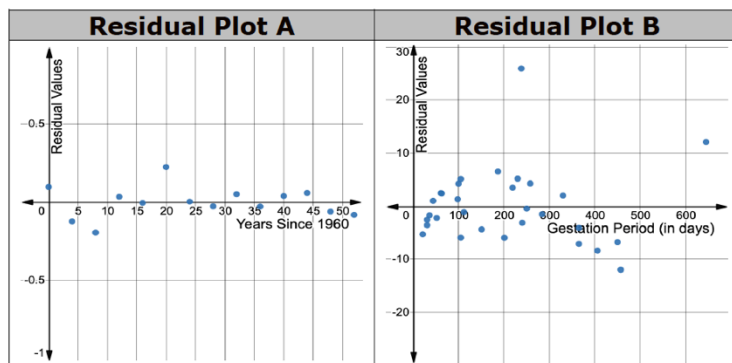
For this activity, students will use the information found in Scatter Plot A and Scatter Plot B to answer a question on a following page. Students may need direction that there are no specific tasks to complete on this page, but that this page will be necessary to complete the question on the following page. Students may not initially recognize the scatter plots here are the same used in the previous two components.

3.8.4 Bring It Together: Interpreting the Strength and Direction of a Line of Fit (continued)

1. Explain which data appears to have a stronger linear association: the data represented on scatter plot A or scatter plot B?

Answers will vary. The data represented in Scatter Plot A appears to have a stronger linear association because the points are all very close to the line of best fit. The data in Scatter Plot B is more spread out around the line of fit, so while it shows a linear association it is not as strong. Scatter Plot B also shows outliers that weaken the linear association.

2. The residual plots for both scatter plots are shown.



Explain how the residual values shown on the residual plots above support your answer to question 1.

Answers will vary. The residual values for the data represented on Scatter Plot A are very small. All of them are between -0.5 and 0.5, which means the line is a good fit and there is a stronger linear association. The residuals are scattered around $y=0$ showing that a line is a good fit for the data. The residual values of the data represented on Scatter Plot B are larger and show a few that are significantly different from the line modeling the relationship. This suggests the fit of the linear function is not as good.



The language around this concept is very important. Consider having students first discuss and share out their responses to question 1, so that they can hear different and diverse perspectives and methods. Use informal observation from the previous two components to dictate how much guidance to provide students in writing this explanation.



Why does a residual plot look different than the original scatter plots?



Have students share out their answers to question 2, connecting it to their responses in question 1.

3.8.4 Bring It Together: Interpreting the Strength and Direction of a Line of Fit (continued)

3. The linear functions fit to each set of data are shown.

Linear Function Fit to Scatter Plot A	Linear Function Fit to Scatter Plot B
$f(x) = 10.2217 - 0.0098x$	$g(x) = 0.032x + 7.6$

Complete the statements.

The slope of the function fit to scatter plot A is

- ☐ negative,
- ☐ positive,

which means the finishing time for

Olympic gold medalists in the men's 100-m dash

generally ☐ increases ☐ decreases as the number of years since

1960 increases. The strength of this linear relationship

is ☐ strong ☐ weak because the residual values are

- ☐ relatively small
- ☐ relatively large

and

- ☐ randomly scattered around the x -axis.
- ☐ show a pattern.

The slope of the function fit to scatter plot B is

- ☐ negative,
- ☐ positive,

which means the longevity of an

animal's life generally ☐ increases ☐ decreases as the gestation period increases.

The strength of this linear relationship is ☐ strong ☐ weak

because the residual values are

- ☐ randomly scattered around the x -axis.
- ☐ show a pattern.

- ☐ relatively small
- ☐ relatively large

and



Consider breaking this into two components: Scatter Plot A and Scatter Plot B. Have half the class work on Scatter Plot A, and the other half work on Scatter Plot B. Then, read the statements aloud, pausing to allow the half of the class focused on that scatter plot to respond aloud with the correct response. If there are any places where there is disagreement about the answer, call on students to defend their answer and share additional ideas until the group reaches consensus.

3.8.5 Wrap-Up: Reflection

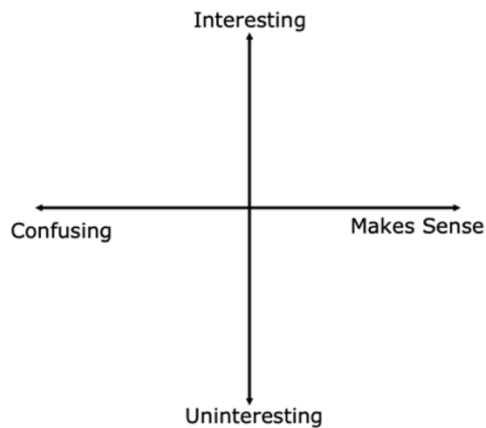
Component Overview: Students complete a self-reflection on their level of understanding of the content of this lesson.



3.8.5 Wrap-Up: Reflection

- Plot a point on the grid below to indicate how you feel about this lesson's learning targets.
 - ☐ I can determine whether a function is an appropriate fit for the data by analyzing residuals.
 - ☐ I can interpret the strength and direction of the correlation within a real-world context.

Answers will vary.



What would reflection look like for you as a teacher? Consider your own scale for strengths and gaps for students through the lesson, and how best to address potential misconceptions or gaps in understanding prior to the unit assessment.